IMPERIAL

TME Mediated MTA Metabolism Confers Resistance to MAT2A Inhibition and induces Multi-Omic Reprogramming in MTAP-Deficient Cancer

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Background

-MTAP deletions occur in approximately 15% of all human cancers.

--MTAP-loss may result in methylthioadenosine (MTA) accumulation, an inhibitor of protein arginine methyltransferase 5 (PRMT5).

-MTAP-deficient cells become dependent on MAT2A in the methionine cycle to maintain intracellular SAM and methionine.

-While MAT2A inhibition (MAT2Ai) has demonstrated success *in vitro*, clinical translation, evident by the AG-270 monotherapy trial, has been limited by modest therapeutic benefit and metabolic resistance.

Hypothesis and Objectives

Hypothesis: **A resistant clinical phenotype arises from the tumor microenvironment (TME); Protein methylation influences resistance to MAT2Ai in MTAP$^{-/-}$ Cancer.**

We aim to:

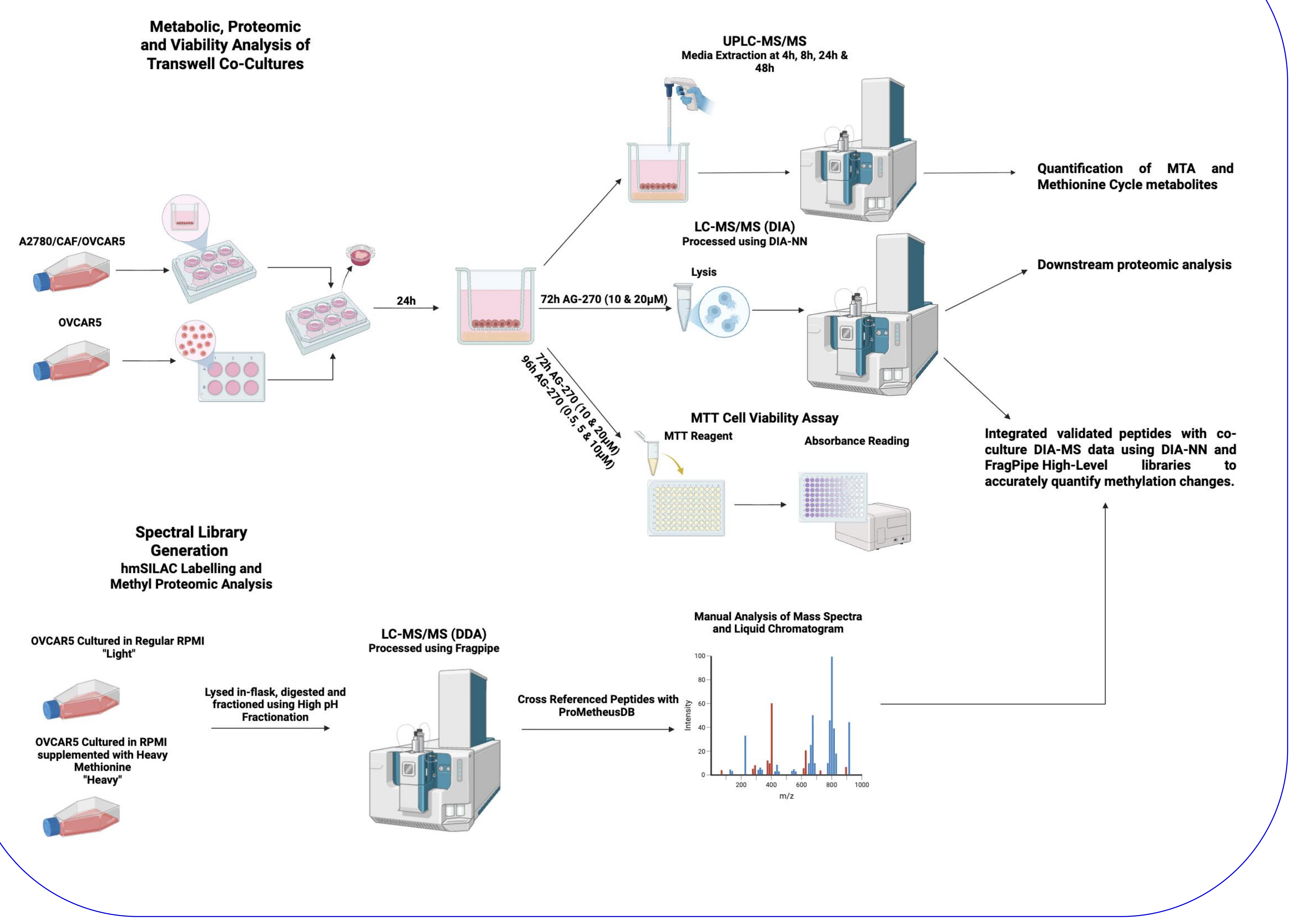
-Assess changes in extracellular MTA.

-Determine whether co-cultures with MTAP-competent cells reduces sensitivity to AG-270.

-Characterize proteomic changes induced by the TME-mediated resistance.

-Characterize proteomic and methylation changes in an MTAP-competent TME under MAT2Ai.

Methods



Results

The *MTAP*$^{+/+}$ Microenvironment Confers Resistance Towards MAT2A Inhibition and clears excess MTA

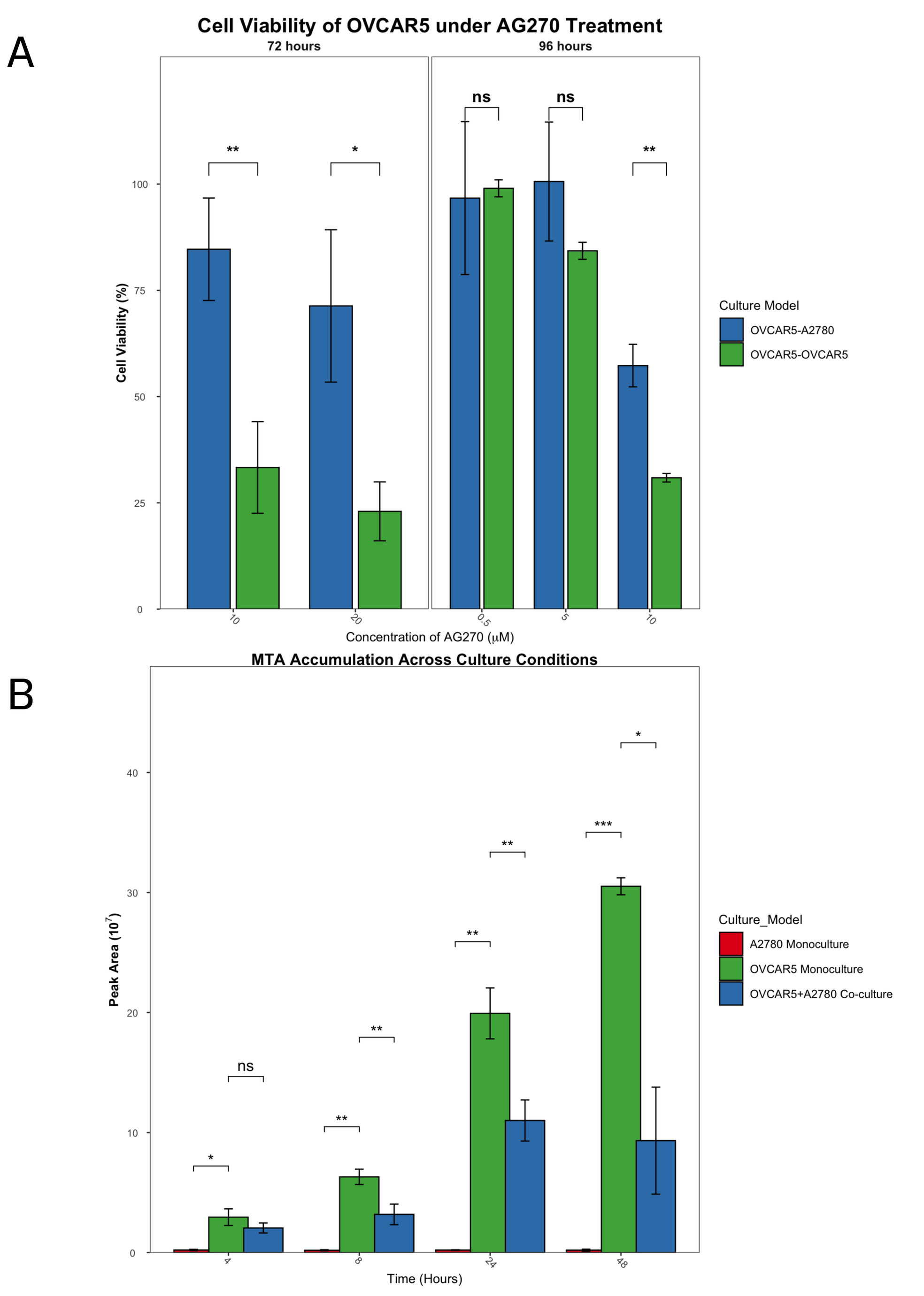


Figure 1. (A) Cell viability assay demonstrates that OVCAR5 (MTAP-/-) cells exhibit significantly higher resistance to AG-270 treatment when co-cultured with MTAP+/+ A2780 cells compared to OVCAR5 monocultures, across 72h and 96h time points. (B) Extracellular metabolomic analysis reveals that A2780 co-culture models significantly reduce the accumulation of MTA in the culture medium compared to OVCAR5 monocultures, indicating active TME-mediated metabolic siphoning.

The *MTAP*$^{+/+}$ TME Induces Proteomic Reprogramming of *MTAP*-deficient cells

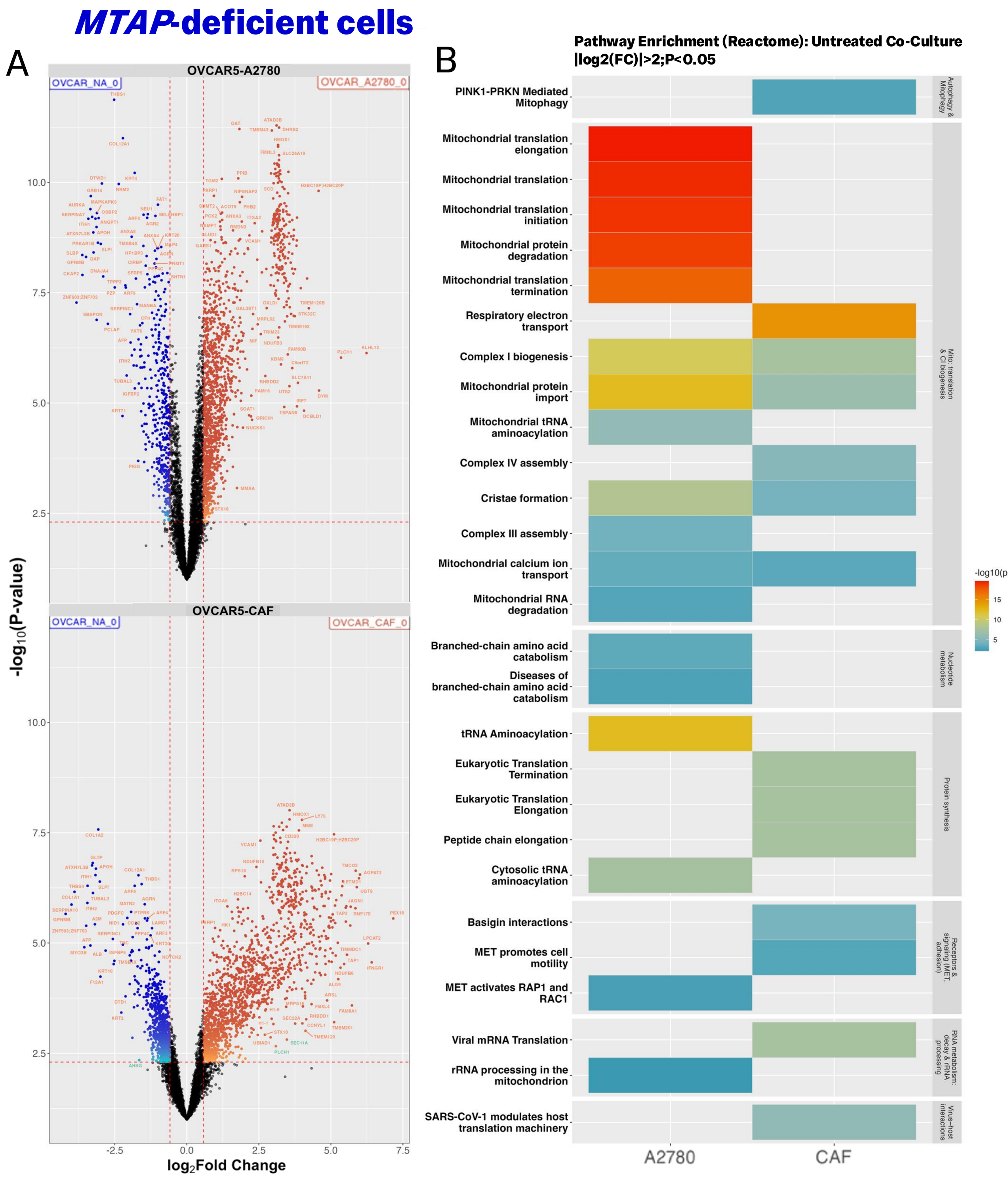


Figure 2. (A) Volcano plots illustrating the differential protein expression in OVCAR5 cells co-cultured with MTAP-competent A2780 cells (top) or CAFs (bottom) compared to OVCAR5-OVCAR5 control. (B) Reactome pathway enrichment heatmap highlighting significantly enriched pathways in co-culture models. MTAP-competent co-cultures exhibit a coordinated survival-oriented reprogramming characterized by the upregulation of mitochondrial translation, CI biogenesis, and protein synthesis, contrasting with the stress-related pathways enriched in isolated MTAP-deficient cultures.

The *MTAP*$^{+/+}$ TME buffers the Proteomic and Methyl-Proteomic Collapse induced by MAT2Ai

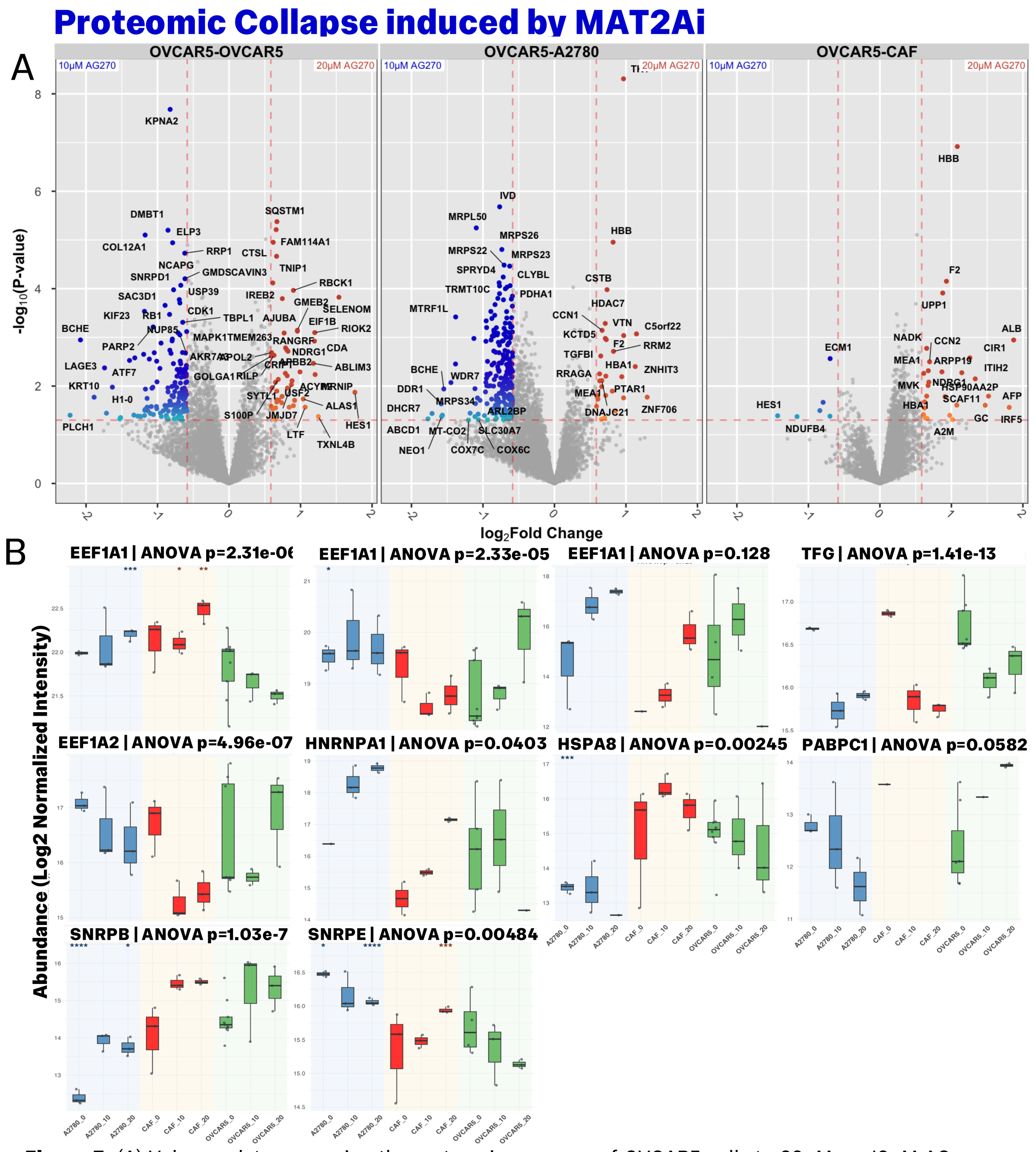


Figure 3. (A) Volcano plots comparing the proteomic response of OVCAR5 cells to 20µM vs. 10µM AG-270. MTAP-deficient OVCAR5-OVCAR5 co-cultures (197 proteins) exhibit widespread proteomic stress characterized by autophagy, lysosomal proteolysis, and ER stress, indicating a proteotoxic crisis. In contrast, OVCAR5-A2780 co-cultures (260) exhibit a coordinated survival-oriented metabolic reprogramming, while OVCAR5-CAF co-cultures (33 proteins) display a highly stabilized proteome with minimal differential expression. (B) Methyl-proteomic profiling using validated peptides from hmSILAC experiments reveal methylation changes not limited to PRMT5-mediated reactions. Stromal buffering prevents the widespread proteomic collapse observed in *MTAP*-deficient co-cultures, allowing cancer cells to maintain homeostatic machinery.

Discussion

-MTAP-competent co-culture reduces sensitivity to MAT2Ai.

-MTAP-competent stromal cells reduce extracellular MTA.

-MTAP-competent co-culture induced changes to protein methylation.

-Methylation changes were not limited to PRMT5-mediated methylation

-Proteomic data suggest mitochondrial autophagy may help stabilize cells in response to MAT2Ai.

-These findings support the TME's role in mediating resistance to MAT2A inhibition.

Limitations

-Transwell models do not fully capture the physical heterogeneity or immune-modulatory complexity of the in vivo TME.

-The specific signaling molecules (beyond MTA) that trigger the pre-emptive upregulation of mitochondrial import machinery in OVCAR5 cells remain unresolved.

Future Prospects

-MAT2A inhibitor monotherapy may be insufficient in heterogeneous tumors.

-Targeting tumor-stroma interactions may enhance MAT2A inhibitor efficacy.

-Promising therapeutic targets include:

- Serine biosynthesis pathways
- FABP4-mediated lipid metabolism
- Other stromal-mediated metabolic interactions

-Further work is required to fully characterize the methyl-proteome, and the possible effects on the epigenome.

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